MILITARY SPECIFICATION

Accelerometer, Electrical, Linear
ABU-13/A

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall operate as an acceleration sensor and electronic voltage multiplier.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
ABU-13/A	3.5

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Case, Base, Mounting, and Mounts, Vibration For Use With Electronic Equipment in Aircraft
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

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MIL-T-18303 Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for

MIL-N-18307 Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment.

Naval Air Systems Command

AR-5 Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of

AR-8 Versatile Avionics Shop Test System, Avionics Systems Compatibility, General Requirements for

AR-9 Versatile Avionics Shop Test System Test Program, General Requirements for

AR-10 Maintainability of Avionics Equipment and Systems, General Requirements for

STANDARDS

Military

MIL-STD-461 Electromagnetic Interference Characteristics, Requirements for Equipment

MIL-STD-462 Electromagnetic Interference Characteristics, Measurement of

MIL-STD-463 Definitions and Systems of Units, Electromagnetic Interference Technology

MIL-STD-704 Electric Power, Aircraft, Characteristics and Utilization of

MIL-STD-781 Reliability Tests: Expotential Distribution

MIL-STD-785 Requirements for Reliability Program for Systems and Equipments

MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of

FED-STD-595 Federal Standard Colors

PUBLICATIONS

Naval Air Systems Command

EI-529 Avionics Installation Instructions for Accelerometer, Electrical, Linear

2.1.1 Availability of Documents -

(1) When requesting specifications, standards, drawings and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection

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with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.5.)

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5.

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3.2.3 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with Specification AR-10.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed two pounds.

3.3.2 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 1,000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 8,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment, including any Built-in-Test provisions shall have 1,000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements 4.4.3.

3.3.2.4 Shelf Life - The equipment shall remain serviceable without maintenance when kept in storage for five years. The reliable operating life shall not be adversely affected by such storage. Shelf life includes those periods of time in which the unit is stored:

(1) On the shelf.

(2) During shipping.

(3) In an aircraft which is in a non-operating status.

3.3.3 Cabling and Connections -

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3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.3.3 Receptacle Orientation - Receptacles mounted on vertical surfaces shall be so oriented that the receptacle keyway is at the bottom of the shell of the receptacle.

3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461, MIL-STD-462 and MIL-STD-463.

3.3.6 Provisions for Maintainability - The Maintainability Program, built-in test features, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in Specification AR-10.

3.3.6.1 Compatability With VAST - The equipment shall be compatible with the Versatile Avionic Shop Test System (VAST) and shall meet the requirements of Specification AR-8. Sufficient test points shall be provided, in accordance with AR-8 and AR-10, to allow an end-to-end test and for fault isolation to either the electronic multiplier unit or the sensor. Fault isolation to a level within the electronic multiplier is not a requirement. When required by contract, VAST Test Programs shall be furnished in accordance with Specification AR-9. If VAST Test Programs exist

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for the equipment, and changes to the equipment are made which will affect the fault diagnosis procedure, changes to the existing test program shall be prepared as part of the equipment changes in accordance with Specification AR-9.

3.3.6.2 Self Test Feature - The self test feature shall provide an indication of satisfactory operation when installed in the aircraft.

3.3.6.3 Field Maintenance - The equipment requires no field maintenance, calibrations or adjustments. The unit is not repairable in the field and must be returned to the factory for service.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room ambient (25°C ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 V AC

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein:

3.3.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400.

3.3.9.2 Temperature - The equipment shall operate satisfactorily

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and meet all performance requirements specified in Specification MIL-E-5400 except that the temperature extremes shall be -54°C and +100°C.

3.3.9.3 Shock - The equipment shall operate satisfactorily and shall not suffer damage after being subjected to the shock requirements of Specification MIL-E-5400 except that the impact G level shall be 25G.

3.3.9.4 Acceleration (Steady State) - The Accelerometer, Electrical, Linear must meet all operational requirements when subjected to steady state accelerations up to 6G's in any direction along any axis.

3.3.9.5 Design Load Factor - When mounted directly to a mounting panel, the article shall be capable of withstanding a continuous load factor of 7G's caused by aircraft accelerations in both directions along each of three mutually perpendicular axes. Permanent degradation in performance of structural yielding shall not occur below 115% of the design load factor. Structural failure shall not occur below 150% of the design load factor.

3.3.9.6 Undervoltage Protection - The article shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within specified limits.

3.3.10 Warm-up Time - The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed two minutes under standard conditions and five minutes at extreme service conditions.

3.3.11 Input Electrical Power -

3.3.11.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704 and limits as specified below. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase), 115V, Category B, 6 VA.

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3.4 Performance - Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall operate as an acceleration sensing transducer and analog voltage multiplier which will be mounted on an aircraft framing member without shock mounts. Installed in the aircraft, its sensitive axis will coincide with the aircraft vertical and its operating range will cover upward accelerations from 0 to +6G's. The CP-741(V)/A computer provides a signal input to the multiplier which may range from 0 to 10 volts DC. This input will be referred to as the primary input. The primary input may have an 800 Hz ripple of up to 100 mv (RMS) superimposed on the DC voltage. The source of the primary output of the Accelerometer is a magnetic amplifier. The primary output from the Accelerometer shall be a direct current voltage varying as a function of input acceleration and primary input voltage and is the input signal to an integrator in the CP-741(V)/A Weapons Release Computer. The secondary output from the Accelerometer shall be a direct current voltage varying as a function of input acceleration.

3.4.2 Self Test - A self-test mode of operation is required to enable system testing or component testing. When in the self-test mode of operation, simulated accelerations of 1.4, 3.5 and 5.0G units shall be provided individually to the Accelerometer. The simulated acceleration input is required to physically exercise the acceleration sensing transducer. Also, a self-test reference direct current voltage shall be available from the Accelerometer.

3.5 Detail Requirements -

3.5.1 Accelerometer, Electrical, Linear ABU-13/A - The Accelerometer shall meet the following requirements:

3.5.1.1 Function - The Accelerometer shall provide the following

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functions:

(1) A function proportional to the ratio of the normal acceleration of an aircraft to the full-scale acceleration (6G) of the Accelerometer and is used to provide information to a CP-741(V)/A Weapons Release Computer.

(2) A function proportional to the normal acceleration of an aircraft and is used to provide information for a programmed "pull-up" maneuver.

3.5.1.2 Form Factor - The Accelerometer shall be enclosed in a case conforming to the general requirements of MIL-C-172. The outline dimensions of the case shall conform to EI-529.

3.5.1.3 Weight - The weight of the Accelerometer shall not exceed two pounds.

3.5.1.4 Case - The case of the Accelerometer shall be made of non-ferrous, low density metal, uniform in texture and shall have a smooth surface. The case shall be finished with a lusterless gray material, color number 35237 of FED-STD-595. The sensitive axis shall be marked on the case as shown in EI-529.

3.5.1.4.1 Hermetic Sealing - The case shall provide a hermetically sealed enclosure for all of the mechanism.

3.5.1.5 Mass CG Location - The location of the mass center of gravity of the sensing element of the Accelerometer will be clearly marked on the case.

3.5.1.6 Installation Mounting - Installation shall be with the sensitive axis vertical. Acceleration in an upward direction shall be defined as position (+G's).

3.5.1.7 Alignment - The sensitive axis of the Accelerometer shall be perpendicular to within one degree of the line connecting the centers of the two mounting holes at the connector end of the case.

3.5.1.8 Electrical Connections - Connections to external circuits shall be provided in accordance with EI-529, Dwg. EI-529-2, and shall be clearly shown on the outside of the case.

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3.5.1.9 Range - The Accelerometer output shall be within specified tolerances over the following range of input conditions:

Input acceleration from $0^{+0.200}_{-0.000}$ units to $6^{+0.000}_{-0.200}$G units

Primary input voltage 0.0 VDC to 10.0 VDC

3.5.1.10 Limits on Voltage Output - For input accelerations along the sensitive axis of the Accelerometer exceeding 6G, the primary output voltage E_o shall not exceed the output voltage corresponding to 6G units.

3.5.1.11 Short Circuit Protection - All outputs of the article when operating shall be suitably protected so that any output pin can be shorted to signal common for extended periods of time without causing permanent damage to the article.

3.5.1.12 Input Impedance - Primary input impedance of the Accelerometer shall not be less than 2K ohms.

3.5.1.13 Output Impedance - The output impedance of either the primary output or the secondary output shall not be greater than 5K ohms. The output impedance of the self-test reference direct current voltage source shall not be less than 2.4K ohms or greater than 4.8K ohms.

3.5.1.14 Performance Characteristics -

3.5.1.14.1 Primary Output Accuracy - Primary output voltage E_o is defined as follows:

$$E_o = \frac{E_i A_s}{6G}$$

The error in the steady state primary output voltage is defined as follows:

$$E_{o_\epsilon} = E_o - \frac{E_i A_i}{6G}$$

where E_o = primary output voltage

E_i = primary input voltage

A_i = input acceleration along the sensitive axis of the Accelerometer, in G units

A_s = acceleration sensed by the Accelerometer

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E_{o_ε} = primary output voltage error

G = acceleration units

The maximum allowable primary output error $\varepsilon_{o_{max}}$ is as follows:

$$\varepsilon_{o_{max}} = \frac{E_{o_\varepsilon}}{E_i} \times 100\% \leq \pm .67\% \text{ from } -54°C \text{ to } 70°C$$

or ± 20 mv whichever is the larger

$$\varepsilon_{o_{max}} = \frac{E_{o_\varepsilon}}{E_i} \times 100\% \leq \pm 1.00\% \text{ from } 70°C \text{ to } 100°C$$

or ± 30 mv whichever is the larger

Figure 1 shows the above error limits in graphical form.

3.5.1.14.2 Secondary Output Accuracy - The secondary output voltage E_s is defined as follows:

$$E_s = K_1 - K_2 A_i$$

where E_s = secondary output voltage

A_i = input acceleration along the sensitive axis of the Accelerometer, in G units

K_1 = 17.5 volts

K_2 = 2.8 volts/G

The maximum allowable error in the secondary output voltage is as follows:

$E_{s_{\varepsilon max}} = K_2 A_\varepsilon$ when the secondary output load impedance is 150K

where A_ε = ± .06G units from -54°C to 70°C

A_ε = ± .09G units from 70°C to 100°C

3.5.1.14.3 Self Test Accuracy - The accuracy of the test simulated acceleration input shall be ± 0.06G units from -54°C to 70°C, and ± 0.09G units from +70°C to +100°C.

3.5.1.14.4 Dynamic Response - The response of the primary output voltage shall meet the conditions of paragraphs 3.5.1.14.5, 3.5.1.14.6 and 3.5.1.14.7 with any combination of the following input conditions:

E_i = +1 VDC to +10 VDC

A_i = +1G step to +6 step

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3.5.1.14.5 Rise Time - The primary output voltage shall rise to at least 63% of the specified steady state voltage in 10 ms or less.

3.5.1.14.6 Overshoot - The primary output voltage shall have less than 5% overshoot.

3.5.1.14.7 Settling Time - The primary output voltage shall settle to specified steady state voltage in 100 ms or less.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified, the supplier is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test production of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a Government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor

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shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on the equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of tests - This equipment may be subjected to any

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and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitive results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

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(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run in Test - Each equipment shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 CPS (excluding resonant points) and a minimum amplitude of ± 3G's

The equipment shall be vibrated (without vibration isolators) for a period

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of ten minutes prior to the beginning of the ten-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the equipment in two directions, the vertical direction shall be used. During the ten-hour period of operation following the ten-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The ten-hour period specified above may be composed of two five-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Sampling Tests are not required when Reliability Assurance Tests are conducted.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions,

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making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan III shall be used.

4.4.3.2. Reliability Production Acceptance (Sampling) Phase Tests - The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". Test Level E shall be used.

4.4.3.2.1 All Equipment Test - Each equipment produced except those submitted for the Reliability Qualification Test, shall be tested for thirty hours. Prior to the thirty-hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to

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be used, the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month, the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required.

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(This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case of design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
AC Voltage	115 ± 5 volts (at lowest applicable frequency)

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient furation to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately eight-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer

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(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted

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for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - The equipment covered by this specification is intended for use in aircraft to provide:

(1) A function proportional to aircraft angular rate to a CP-741(V)/A Weapons Release Computer.

(2) A function proportional to the normal acceleration of an aircraft and is used to provide information for a programmed "pull-up" maneuver.

6.2 Ordering Data - Purchases should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number and date of this specification

(2) Selection of applicable levels of packaging and packing (See 5.1)

6.3 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from

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subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Non-Repairable Subassemblies - As a general rule, non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-To-Failure greater than 5,000 hours, and for many applications, this figure must be nearer 50,000 hours.

6.6 Type Designations - The parentheses () when used in the equipment type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The

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complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Revisions - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Associated Equipment -

Computer, Weapons Release, CP-741(V)/A

Bomb Display Coupler, CU-1466

Preparing Activity
Navy - AS

Project Number 1260-NO14

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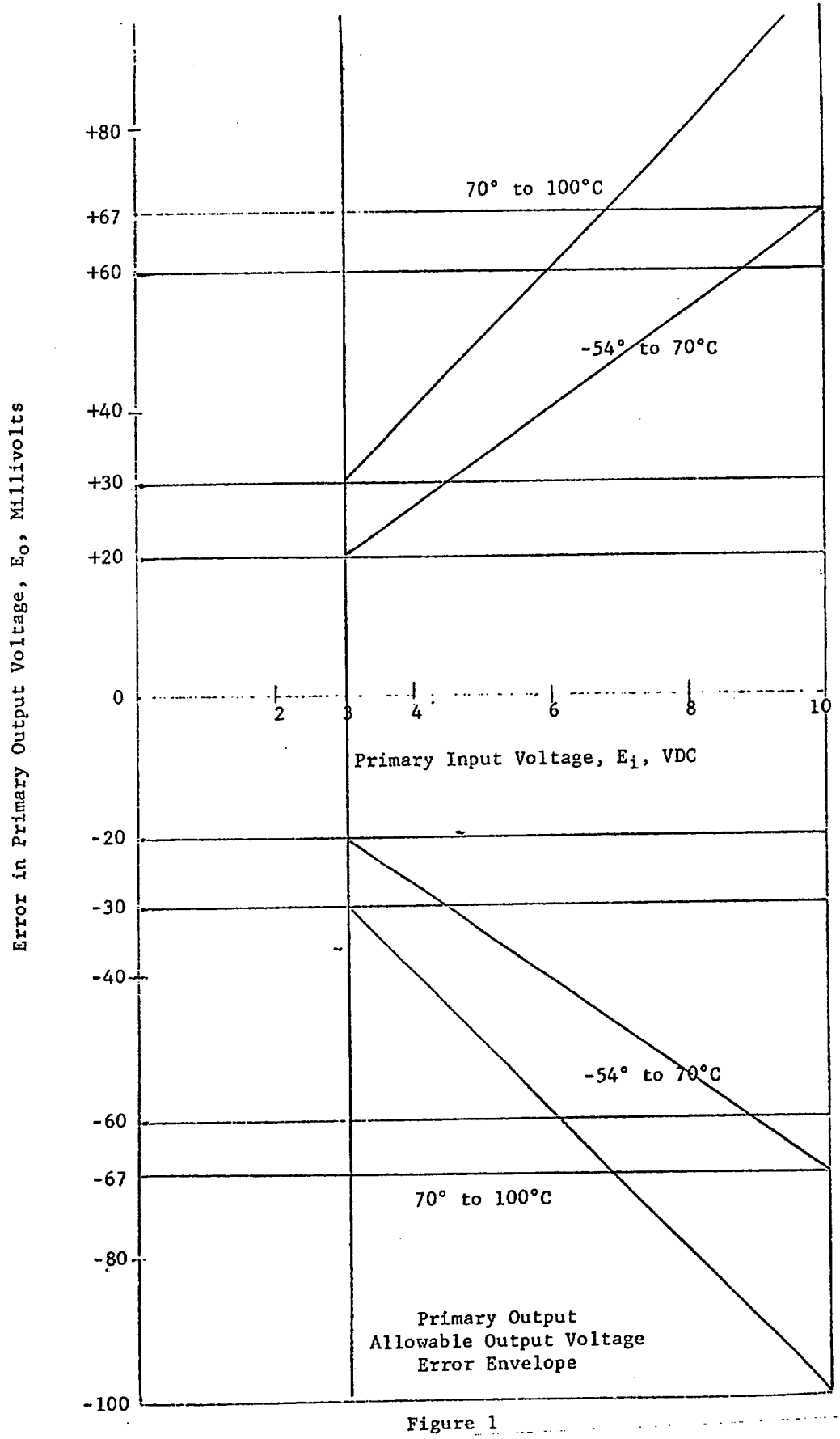


Figure 1

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